

**The Mass Gap and Self-Interaction in Flowing Space:
Discreteness, Noncommutativity, and a Lattice Yang–Mills
Structure**

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Abstract

The Yang–Mills existence and mass-gap problem [1] asks for a quantum field theory of a non-abelian gauge group that is defined rigorously and exhibits a positive mass gap. This paper does not solve that problem. It derives, in the algebraic setting of a flowing-space framework, the two structural ingredients that the problem names — a discrete mass spectrum with a positive lower bound, and a non-abelian self-interaction — and shows that in this framework they are the same mechanism viewed twice. The argument has four steps. (i) *Discreteness*: mass is the number of strands of excited flow; the square of the mass is an integer multiple of a unit M_0^2 (MG1). A glueball witness makes the discreteness concrete: the repository’s glueball mass squared is an integer, either 0 or at least 1 (MG4). The transition from the electron ($N \geq 1$ strands) to the photon ($N = 0$) is an integer jump that cannot be crossed continuously (MG3); hence the interval $(0, M_0^2)$ is empty (MG2b). (ii) *Self-interaction*: the sink contraction of negative charge is a self-flattening dynamics — iterating it reduces the fluctuation energy by $(1 - \lambda)^{2n}$ (YM1) — and two sink points act noncommutatively: $T_p T_q \neq T_q T_p$ (YM2), with the commutator exactly $\lambda^2(v_q - v_p)$ at p (YM3) and magnitude $\lambda^2|v_q - v_p|$ (YM3b). (iii) *Lattice gauge field*: on a $\mathbb{Z}$ lattice, a color-matrix-valued field A has field strength $F(t) = [A(t), A(t+1)]$, the nearest-neighbor commutator (CA3); $F \neq 0$ iff the neighbors do not commute (CA4), which is the lattice avatar of $F = \partial A - \partial A + [A, A]$ [2]. (iv) *Continuum limit mechanism*: the framework’s own derivation of the four forces — differentiating the flow momentum $m(C-v)$ — is adopted as the existence mechanism for the continuum limit (CA8–CA10), and the strict estimates L_1 – L_3 bound the commutator norm by the explicit factor 6 and prove the nuclear channel norm-nonzero. The assembled chain closes conditionally: accelerated charge, the self-flattening sink, the noncommutative self-interaction, the continuum mechanism, and the mass gap are one conditional sequence (CA7, CA10). All statements are formalized in Lean 4 with zero `sorry`; the honest boundaries — the discrete-to-continuous topological limit remains open — are stated rather than blurred.

“Water is the origin of all things.”

— *Thales of Miletus*

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I. INTRODUCTION

a. The two facts carried by the problem's name. The Yang–Mills mass-gap problem [1] is the question whether a non-abelian gauge theory can be constructed rigorously and whether it has a positive mass gap. Two structural facts carry its physical content. The first is *discreteness*: the spectrum of a confining theory has no massless excitation — the lightest glueball sits at about 1.6–1.7 GeV, far above the vacuum [3–5]. The second is *self-interaction*: the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \quad (1)$$

is nonlinear, the field couples to itself through the commutator [2]. In the standard framework the two facts are unrelated and neither is derived: gauge invariance *postulates* $[A, A]$, and the gap is not a theorem but a non-perturbative measurement [1, 6]. This paper's claim is that in the flowing-space framework they are one mechanism, and both are theorems.

b. The claim. Within the framework's four postulates — space is a flowing medium; mass is an integer count of excited strands, $m^2(N) = NM_0^2$; charge is the divergence of the flow; the four forces are the Leibniz decomposition of the flow momentum $m(C - v)$ — the following are proved in Lean 4 with zero `sorry` [7, 8]:

- *The mass gap is integer arithmetic.* Because N is an integer, the electron–photon transition is a jump $N \geq 1 \rightarrow N = 0$, and the interval $(0, M_0^2)$ is empty (MG1–MG4). The discreteness of a count, not an assumption about confinement, produces the gap.
- *The self-interaction is sink noncommutativity.* Two sink contractions at distinct points do not commute; their commutator is exactly $\lambda^2(v_q - v_p)$, the same shape as $[A, A]$ — field-dependent and noncommutative (YM1–YM3b).
- *The field strength is a nearest-neighbor commutator.* On a $\mathbb{Z}$ lattice, a color-matrix-valued gauge field has $F(t) = [A(t), A(t+1)]$, and $F \neq 0$ iff the neighbors do not commute (CA3–CA6).
- *The continuum exists by the four-force derivative.* The framework's own derivation of the forces is adopted as the existence mechanism for the continuum limit, with strict norm bounds (CA8–CA10, L1–L3).

The assembled chain — accelerated charge, self-flattening sink, noncommutative self-interaction, continuum mechanism, mass gap — is a conditional sequence of theorems (CA7, CA10).

c. What this paper is and is not. It is not a claim to have solved the Yang–Mills problem: the lattice construction is an algebraic skeleton, the topological limit of the lattice spacing is not formalized, and the numerical inputs λ , M_0 are not derived. These boundaries, and the conditions under which the program dies, are stated in Sec. IX. What the paper does claim is narrower and verifiable: the two names of the problem are one mechanism, and each step is a machine-checked theorem.

d. How to read this paper. Sections II–IV carry the central claim (postulates; mass gap from discreteness; self-interaction from noncommutativity) and suffice for the reader who wants it. Sections V–VII give the lattice mechanics and its strict estimates; Sec. VIII assembles the full chain; Secs. IX–X assess honestly and list the open problems. The motivation — how the framework grew from the Riemann hypothesis — is told in Sec. II; the dialogue with existing physics is in Sec. VIII.

II. POSTULATES OF THE FLOWING-SPACE FRAMEWORK

We recall the framework’s postulates, in the form used by the preceding paper and its Lean formalization. Each postulate is stated, then read physically, so that the geometry of Secs. III–VIII remains anchored.

Postulate II.1 (Flowing space). *Space is a flowing medium. A material point comoving with the flow moves at the vector speed C ; the flow momentum is $p = m(C - v)$, where m is the mass and v the velocity of the point relative to an observer.*

Reading. The epigraph of this paper is not decoration. Thales’ “water is the origin of all things” is the first recorded statement of the primitive that this framework makes precise: the substance of physics is a medium in motion, not a void with fields in it. The river model of black holes [9] is the same picture made quantitative — space flows inward through the horizon, and the speed of light is the speed of the medium. The momentum $p = m(C - v)$ is then the natural bookkeeping: a body that is carried completely by the flow ($v = C$) has zero flow momentum (the photon), while a body that anchors against the flow ($v \neq C$) carries momentum proportional to the relative speed.

Postulate II.2 (Mass is strand count). *The mass squared of an excited configuration is an integer multiple of a fundamental unit:*

$$m^2(N) = N M_0^2, \quad N \in \mathbb{N}, \quad (2)$$

where N is the number of strands of excited flow and M_0 is the strand mass. The photon has $N = 0$ and $m = 0$.

Reading. The picture is a boat moored to the bank of a river: each mooring rope is a strand of excited flow, and the more strands grip the flow, the harder it is to tear the pattern free. The mass squared counts the strands — $m^2(N) = N M_0^2$ — so the spectrum is discrete by construction: $M_0^2, 2M_0^2, 3M_0^2, \dots$. The photon has no moorings at all: $N = 0$, it is carried completely by the flow, massless and neutral. The strand count is the same integer N that appears throughout the repository as the glueball configuration energy; the counting postulate is the algebraic seed of every discreteness statement in this paper.

Postulate II.3 (Charge is divergence). *Positive charge is a source of the flow (positive divergence); negative charge is a sink (negative divergence); neutral is zero.*

Reading. The picture is a faucet and a drain. A positive charge is a faucet: displacement flows outward from it. A negative charge is a drain: flow converges into it. The two are geometric opposites — flipped under charge conjugation $\rho \rightarrow -\rho$ — and the sign of the divergence is the sign of the charge. In the preceding paper this postulate was turned into theorems: the sink contraction reduces fluctuation energy by a square factor, and a fully contracted sink is flattened, massless, and neutral — the photon. The present paper uses the same two faces of the drain: its self-flattening dynamics (Sec. IV) and its noncommutativity with a second drain.

Postulate II.4 (Four forces are the Leibniz decomposition). *Differentiating the flow momentum $p = m(C - v)$ with respect to time yields the four channels*

$$\dot{p} = \dot{m} C + m \dot{C} - \dot{m} v - m \dot{v}, \quad (3)$$

identified respectively with the electric, nuclear, magnetic, and gravitational (inertial) forces.

Reading. This is the framework’s answer to “where do the forces come from”: they are the four terms of a Leibniz decomposition, the change rates of the flow bookkeeping. A change

in the mass (anchoring) couples to the flow speed: $\dot{m}C$, the electric channel. A change in the flow itself: $m\dot{C}$, the nuclear channel. A change in mass coupling to the body velocity: $-\dot{m}v$, the magnetic channel. A change in the body velocity: $-m\dot{v}$, the inertial channel. The postulate matters for this paper in one specific way: the four-force derivative is the framework’s *built-in continuum*, the existence mechanism for the continuum limit (Sec. VI).

The four postulates are the entire physical input. Everything below is derived from them in Lean 4.

a. Motivation: how the chain grew. The motivation is cumulative, and each of its earlier forms was abandoned for a concrete reason — the same pattern as the preceding paper of the series [8, 10]. It began with the Riemann hypothesis [11]: the critical line $\text{Re } s = 1/2$ is the fixed set of the reflection $s \mapsto 1 - s$, and a value of $1/2$ forced by a symmetry, rather than put in by hand, seemed too physical to be a coincidence. That led to an attempt to build a “hidden-number space” on the critical sheet; the attempt was abandoned when a single sheet could not accommodate the distinction between fermions and bosons — one object, one sheet, no room for two statistics [12]. The next goal was to understand where the glueball mass comes from: the lattice spectrum [3, 4] gives the masses but not their origin. The resolution arrived as a realization rather than a derivation: the failures shared a single cause. Space is not a background in which things happen; it is the medium that happens. In that medium mass is anchoring, charge is divergence, a sink is a converging flow, and a fully contracted sink is a flattened flow — massless and neutral, the photon. Two further steps, both made in the present paper, carry the realization to the Yang–Mills problem. If mass is a count, then the mass spectrum is discrete, and the gap between massless and massive is a theorem of integer arithmetic. If the sink is a contraction, then two sinks do not commute, and the noncommutativity of the count-carrying operators is the self-interaction. Each ingredient of this paper is a theorem that makes one step of that realization precise.

III. THE MASS GAP FROM DISCRETENESS

a. The integer jump. Postulate II.2 makes the mass spectrum discrete by construction: $m^2(N) = NM_0^2$ with $N \in \mathbb{N}$. The physically interesting content is not the discreteness itself but its consequence for transitions. A physical process that changes the strand count (an

excitation created or destroyed) changes N by an integer. In particular the electron–photon transition is $N \geq 1 \rightarrow N = 0$: from any massive configuration to the massless one. Because N is an integer, this transition cannot pass through an intermediate value of m^2 strictly between 0 and M_0^2 .

b. Intuition. The mass spectrum is a ladder whose rungs are $0, M_0^2, 2M_0^2, 3M_0^2, \dots$. There is no rung between 0 and M_0^2 — that is all the gap is. The theorems below make this ladder precise and draw its physical consequences.

Theorem III.1 (MG1: mass is an integer count). *For $N \in \mathbb{N}$ and $M_0 \in \mathbb{R}$ with $0 < M_0$,*

- $N \geq 1$ implies $m^2(N) = NM_0^2 > 0$;
- $N = 0$ implies $m^2(0) = 0$.

Formalized as `mass_gap_basics` in `MassGap.lean`.

Reading. Any excited configuration — however small its strand number — has strictly positive mass squared; the massless state is unique, the photon with zero strands. The unit is explicit: one strand gives $m^2 = M_0^2$, two strands $2M_0^2$, three $3M_0^2$. The content is that positivity is *quantized*: there is no configuration whose mass squared is positive but smaller than M_0^2 , because there is no integer strictly between 0 and 1.

Theorem III.2 (MG2/MG2b: the interval $(0, M_0^2)$ is empty). *The lightest massive excitation is the single strand, $m^2(1) = M_0^2$, and every mass squared lies in $\{0\} \cup [M_0^2, \infty)$. In particular there is no mass in the open interval $(0, M_0^2)$.*

Reading. This is the mass gap in its purest form: the set of mass-squared values is a discrete set with a positive minimum above zero. The lightest massive state is the single strand at $m^2 = M_0^2$; every other massive state is heavier. The gap $(0, M_0^2)$ is empty — there is no arbitrarily light but massive excitation, no way to tune a mass continuously down to zero. This is the algebraic seed of the empirical gap of QCD, where the lightest glueball sits at about 1.6–1.7 GeV while the vacuum is at zero [3, 4]. The framework does not yet predict the number 1.6 GeV; it predicts the *shape*: an empty interval below a minimal mass.

Theorem III.3 (MG3: the electron–photon transition is an integer jump). *A transition from $N \geq 1$ to $N = 0$ cannot be performed continuously: for every N , either $m^2(N) = 0$ or $m^2(1) \leq m^2(N)$. This is the algebraic seed of the mass gap: the gap $(0, M_0^2)$ is a theorem of the discreteness of mass, not an assumption.*

Reading. The electron becomes the photon — mass drops from NM_0^2 ($N \geq 1$) to 0. Because the strand count is an integer, the drop is a jump: the transition cannot pass through any intermediate mass squared, because no such value exists in the spectrum. The transition itself is the proof of the gap: if there were masses in $(0, M_0^2)$, the electron could shed strands one at a time and slide down; it cannot, and the discreteness of the count is the reason. Formalized as `electron_to_photon_is_discrete_jump` in `MassGap.lean`.

Theorem III.4 (MG4: the glueball mass squared is an integer). *For any glueball configuration g of the repository, the pure glue mass squared is an integer: either 0 (no excited strands) or at least 1 (at least one excited strand). Formalized as `glueball_mass_gap_present` in `MassGap.lean`.*

Reading. This theorem connects the abstract ladder to the repository’s own glueball model: the glueball mass squared is the configuration field energy, an integer, and the same discreteness argument applies verbatim — the glueball mass-squared values are $0, 1, 2, 3, \dots$ in units of the strand energy. The gap statement for glueballs is “0 or ≥ 1 ”: there is no glueball with mass squared strictly between 0 and the unit. It is a concrete, integer witness that the mass gap of MG2 is not an artifact of the counting postulate but a property of the repository’s glueball states themselves. The empirical anchor — a lightest glueball at about 1.6 GeV with a gap above the vacuum — is the same shape [3–5]; the framework’s unit is not yet tied to GeV, and that calibration is an open problem (Sec. X).

c. Why this is not circular. The mass gap of the Yang–Mills problem is an existence statement about a continuum theory [1]. Here the discreteness is a postulate of the framework, and the gap follows. The honest status: what is proved is the conditional “mass is an integer count $\Rightarrow$ the interval $(0, M_0^2)$ is empty.” The numerical value of M_0 is an input, not derived — it is the second-input gap already named in the preceding paper. The theorem is no less a theorem for being conditional; what it removes from the problem is not the calibration but the mystery: the gap is integer arithmetic, once mass is a count.

IV. SELF-INTERACTION FROM SINK NONCOMMUTATIVITY

a. The sink as a self-flattening operator. A negative charge is a sink: flow converges inward, like water into a drain. In the discrete setting, a sink contraction moves every point

toward the sink value $\bar{v}$ by a fraction $\lambda \in (0, 1)$,

$$v_i \mapsto (1 - \lambda)v_i + \lambda\bar{v}. \quad (4)$$

The preceding paper proved that this reduces the fluctuation energy Q_A by the square factor $(1 - \lambda)^2$. The new observation is that this operator is *dynamical*: applying it repeatedly is a self-feedback loop — the field changes, which changes the effective sink, which changes the field. This is the discrete avatar of a self-interaction: a system that acts on itself, with no external agency in the loop.

b. Intuition. A drain smooths the surface it swallows, and the smoother surface drains more smoothly: the contraction feeds on its own success. Each application of the sink flattens the profile by a factor $(1 - \lambda)$ in amplitude, hence $(1 - \lambda)^2$ in energy; n applications compound the square.

Theorem IV.1 (YM1: iterated self-flattening). *Let $\text{sinkIter}(A, \lambda, n)$ denote n iterations of the sink contraction on the finite set A . Then*

$$Q_A(\text{sinkIter}(A, \lambda, n)) = (1 - \lambda)^{2n} Q_A(v). \quad (5)$$

The fluctuation energy decays exponentially; the sink flattens itself without external agency.

Reading. The decay is exponential in the number of steps: each step multiplies the energy by $(1 - \lambda)^2 < 1$. The numbers make the mechanism concrete. For a moderate sink, $\lambda = \frac{1}{2}$, ten iterations reduce the fluctuation energy by a factor $(\frac{1}{2})^{20} \approx 10^{-6}$; for a strong sink, $\lambda = 0.9$, ten iterations give $0.1^{20} = 10^{-20}$. This is the discrete avatar of a self-interacting field: the field's own dynamics flattens it, and the flattening accelerates because the effective sink sharpens as the profile flattens. Formalized as `sinkContract_iter_square_decay` in `YangMillsSeed.lean`.

c. Noncommutativity. Two sinks at different points $p \neq q$ define two contractions T_p, T_q . Because each pulls every point toward its own sink value, the order matters.

d. Intuition. Two drains in one basin: if the left drain acts first, the surface is pulled toward the left and then the right drain acts on the already-pulled surface; if the right drain acts first, the result is different. The difference between the two orders is the self-interaction.

Theorem IV.2 (YM2: sink contractions do not commute). *For $p \neq q$ and $0 < \lambda < 1$, there exists a field v such that*

$$T_p(T_q v) \neq T_q(T_p v). \quad (6)$$

The witness: $v_p = 1$, $v_q = 0$; the two compositions differ at p , $1 - \lambda \neq 1 - \lambda + \lambda^2$.

Reading. The witness values are the cleanest possible illustration of the mechanism. Take a field that is 1 at p and 0 at q . Applying T_q first pulls v_p toward $v_q = 0$, giving $1 - \lambda$ at p ; applying T_p first leaves $v_p = 1$ and then T_q pulls it, giving $1 - \lambda + \lambda^2$ — the extra λ^2 records the fact that T_p first moved v_q toward $v_p = 1$. For $\lambda = \frac{1}{2}$ the two values are $\frac{1}{2}$ and $\frac{3}{4}$; for $\lambda = 0.1, 0.9$ and 0.91 . The difference is never zero for $0 < \lambda < 1$. This is the discrete witness of the Yang–Mills commutator: the order of operations on the field matters. Formalized as `sinkToPoint_noncommutative` in `YangMillsSeed.lean`.

Theorem IV.3 (YM3: the commutator is exact). *For any field v and any p, q ,*

$$(T_p T_q v - T_q T_p v)(p) = \lambda^2 (v_q - v_p). \quad (7)$$

The strength of the noncommutative term is the square of the contraction strength times the field gradient. This is the same shape as $[A, A]$ in the Yang–Mills field strength: field-dependent and noncommutative.

Reading. The formula is exact — not an estimate, not a first-order approximation, but an identity proved by algebra. Its structure is the structure of the Yang–Mills nonlinearity: the commutator of the two operators is proportional to the *field difference* between the two points, weighted by λ^2 . Two features deserve emphasis. First, the quadratic weight: the noncommutativity is second order in the contraction strength, so a weak sink ($\lambda = 0.1$) produces a self-interaction of weight 0.01 — weak, but present and exact. Second, the field dependence: where the field is uniform ($v_q = v_p$) the commutator vanishes; the self-interaction lives precisely where the field *varies*, exactly as $[A, A]$ vanishes for a constant abelian field. Formalized as `sinkToPoint_commutator_formula` in `YangMillsSeed.lean`.

Theorem IV.4 (YM3b: noncommutativity grows with the gradient). *For any field v and any p, q ,*

$$|T_p T_q v - T_q T_p v|(p) = \lambda^2 |v_q - v_p|. \quad (8)$$

The magnitude of the noncommutative term is the square of the contraction strength times the magnitude of the field gradient.

Reading. The absolute version sharpens the physical reading: the strength of the self-interaction is controlled by two factors, the sink strength squared (λ^2) and the gradient

between the two sink points ($|v_q - v_p|$). A self-interaction without a field gradient is impossible — the mechanism needs variation to act on, which is the discrete echo of the fact that $[A, A]$ is the non-abelian *curvature* term. Formalized as `commutator_grows_with_gradient` in `YangMillsSeed.lean`.

e. What has been proved. The self-interaction is not imported from gauge theory; it is a theorem about sinks. The commutator formula YM3 is exact in the discrete setting. The identification with $[A, A]$ is at the level of structure (the interpretation layer); the continuum limit of the commutator is part of the open strict program below.

V. LATTICE YANG–MILLS: THE FIELD STRENGTH IS A COMMUTATOR

a. The continuous-to-discrete method. The framework has a standing method for passing from continuous to discrete: continuous derivatives are represented by forward differences on a $\mathbb{Z}$ lattice. The Maxwell equations, in this method, become difference identities. The same method is now applied to a gauge field.

b. Intuition. The forward difference $A(t+1) - A(t)$ is the algebraic seed of the derivative $\partial_t A$: before any limit is taken, it carries the derivative’s content — how much the field changes from one position to the next. The framework’s bridge statement (CA6) is precisely that this seed relation is what the continuous-to-discrete method means. On the lattice, the field strength of a gauge field can therefore only be a combination of differences and commutators of neighboring values.

Postulate V.1 (Lattice gauge field). *A gauge field is a matrix-valued field on the lattice,*

$$A : \mathbb{Z} \rightarrow \text{Mat}_3(\mathbb{C}), \tag{9}$$

where the matrix value lives in the color space of the framework (the 3×3 complex matrices carrying the color cycle C_3).

Reading. The lattice is one-dimensional — position is an integer — and the field value at each site is a 3×3 complex matrix. The dimension 3 is the framework’s color number: the same three that appear as spatial directions, anchoring directions, and colors (the color cycle C_3 , a 3×3 permutation matrix). The field is the carrier of the color structure, and the noncommutativity of matrices is available at every site pair. The forward difference

$$\Delta_1 A(t) = A(t+1) - A(t) \tag{10}$$

is the algebraic seed of $\partial_t A$ (CA2); the commutator

$$[A(t), A(t+1)] = A(t)A(t+1) - A(t+1)A(t) \quad (11)$$

is the seed of the nonlinear term $[A_\mu, A_\nu]$.

Theorem V.1 (CA3/CA4: field strength is the nearest-neighbor commutator). *The lattice field strength*

$$F(t) = [A(t), A(t+1)] = A(t)A(t+1) - A(t+1)A(t) \quad (12)$$

satisfies

$$F(t) \neq 0 \iff [A(t), A(t+1)] \neq 0. \quad (13)$$

A single-point term $[A(t), A(t)]$ is identically zero (the color cycle commutes with itself); the noncommutativity necessarily lives between two distinct positions, exactly as in YM2.

Reading. The field strength of the lattice gauge field is *defined* to be the commutator of neighboring values, and the theorem states the equivalence that carries the physics: the field strength is nonzero exactly when the neighboring colors do not commute. The picture is two hands clasping: the field at one site acts on the field at the next, and each needs the other — a single site clasping itself, $[A(t), A(t)]$, is identically zero. The self-interaction necessarily lives *between* two distinct positions, which is the same geometric fact as YM2: two distinct sinks are required for noncommutativity. Formalized as `FieldStrength` and `field_strength_ne_zero_iff_commutator_ne_zero` in `YangMillsLattice.lean`.

Theorem V.2 (CA5: color-cycle carrier). *When the field is the color cycle C_3 , the single-point field strength vanishes; a nonvanishing field strength requires two distinct color directions. This matches the two-sink structure of YM2.*

Reading. The color cycle is a permutation matrix of order three; it commutes with itself, so a constant field of color-cycle values produces zero field strength everywhere. This is not a failure of the construction but its content: self-interaction requires *variation* — two distinct color directions at two distinct sites — exactly as the commutator YM3 vanishes on a uniform field. A constant color field is the discrete analogue of an abelian configuration: no curvature, no self-coupling.

c. The bridge statement. The continuous-to-discrete bridge (CA6) is the framework's own statement: the lattice difference is the algebraic seed of the continuous derivative. For the gauge field, $F = [A(t), A(t+1)]$ is the seed of the nonlinear term $[A_\mu, A_\nu]$ of the continuum field strength [2, 13]. Formalized as `discrete_to_continuous_bridge_statement` in `YangMillsLattice.lean`; the bridge is a structural correspondence — the analytic limit is handled in the next section and its strict form in Sec. VII.

VI. THE CONTINUUM-LIMIT MECHANISM: FOUR FORCES AS EXISTENCE

a. The mechanism. The framework does not need to *assume* a continuum limit: it already has one. Postulate II.4 derives the four forces by differentiating the flow momentum — that is, by taking the continuous change rates of the flowing medium. This derivation is itself the existence mechanism for the continuum: the continuous field exists because the flow has change rates, and those change rates are the four channels.

b. Intuition. If the medium is real, it has change rates; if it has change rates, there is a continuous description; and the framework has already written that continuous description down — it is the four-force decomposition. The continuum limit is not a separate construction bolted onto the lattice; it is the same derivative that produces the forces.

Theorem VI.1 (CA8: the mechanism is the four-force derivative). *The four-force decomposition*

$$\dot{m}C + m\dot{C} - \dot{m}v - m\dot{v} \tag{14}$$

is the derivative of the flow momentum $m(C - v)$. It is hereby adopted as the existence mechanism for the continuum limit: continuous gauge fields exist because the flowing medium has change rates.

Reading. The theorem is an adoption, and the honesty of the word matters: the four-force derivative already exists in the framework (it is how the forces are derived), and the step taken here is to *name* it as the existence mechanism for the continuum. The content is that the continuum is not an extra ingredient: it is the change-rate structure of the medium. Formalized as `continuous_limit_mechanism_defines_existence` in `YangMillsLattice.lean`.

Theorem VI.2 (CA9: the nuclear channel in the continuum). *In the continuum limit, the lattice field strength corresponds to the nuclear channel $m\dot{C}$: the difference $A(t+1) - A(t)$ becomes $\dot{C}$, and $m \neq 0$, $\dot{C} \neq 0$ imply $m\dot{C} \neq 0$.*

Reading. The bridge from the lattice to the continuum is the identification of the difference with the change rate: the field value $A(t)$ is the continuous field, and its forward difference is the derivative $\dot{C}$ of the flow speed. The theorem then states a fact of ordinary arithmetic: the product of two nonzero quantities is nonzero. The nuclear channel — the framework’s name for the change of the flow itself — is nonzero in the continuum exactly when the lattice self-interaction is present. Formalized as `lattice_commutator_continuous_limit_nuclear` in `YangMillsLattice.lean`.

Theorem VI.3 (CA10: the chain closes). *The continuum mechanism (nuclear channel nonzero) implies the mass gap (CA7), by way of the discreteness of mass (MG2b).*

Reading. This is the closing statement of the paper’s central claim: the self-interaction (Sec. IV), carried into the continuum by the four-force mechanism (CA8–CA9), implies the mass gap (Sec. III). The implication is conditional in the sense made precise throughout: each step is a theorem, and the inputs — λ , M_0 , the flattening power — are named. Formalized as `continuous_limit_mass_gap_chain` and `mass_gap_from_self_interaction` (CA7) in `YangMillsLattice.lean`.

VII. STRICT ESTIMATES

The strict program upgrades the structural correspondence to analytic estimates. The structural theorems of Secs. III–VI identify shapes: commutator, gap, chain. The theorems of this section bound sizes.

a. Intuition. The structural statement “the field strength is the commutator” answers *what* the self-interaction is; the estimates answer *how large* it can be. If the commutator norm were unbounded, the self-interaction could diverge under lattice refinement; the bound below shows it cannot.

Theorem VII.1 (L1: commutator norm bound). *For 3×3 complex matrices with the entrywise sup norm,*

$$\|AB - BA\| \leq 6 \|A\| \|B\|. \quad (15)$$

The lattice field strength is bounded by the field norm: the self-interaction does not diverge under lattice refinement.

Reading. The constant 6 is explicit and has a transparent origin: each entry of the matrix product AB is a sum of three terms (the 3 of the 3×3 matrices), each bounded by $\|A\|\|B\|$, giving $\|AB\| \leq 3\|A\|\|B\|$; the commutator is a difference of two such products, $AB - BA$, so the triangle inequality doubles the bound to 6. The constant is not optimal — the proof is honest about that — but it is enough: the lattice field strength $F(t) = [A(t), A(t+1)]$ is controlled by the product of the field norms at neighboring sites, uniformly in the lattice. Under refinement, the self-interaction cannot blow up. Formalized as `commutator_norm_bounded` in `YangMillsStrict.lean`.

Theorem VII.2 (L2: flow-momentum derivative). *For differentiable $m, C, v : \mathbb{R} \rightarrow \mathbb{R}$,*

$$\frac{d}{dt} [m(t)(C(t) - v(t))] = \dot{m}(C - v) + m(\dot{C} - \dot{v}), \quad (16)$$

the product rule for the four-force channel.

Reading. The four-force decomposition of Postulate II.4 is not a bookkeeping device but a genuine derivative: this theorem verifies, via the product rule, that $\dot{p} = \dot{m}(C - v) + m(\dot{C} - \dot{v})$ is exactly the time derivative of the flow momentum. The four channels are the terms of a real calculus identity. This is the strict version of the continuum mechanism of CA8 — in the scalar setting; the matrix-valued generalization is an open problem (Sec. X). Formalized as `flow_momentum_derivative` in `YangMillsStrict.lean`.

Theorem VII.3 (L3: nuclear channel norm-nonzero). *For $m \neq 0$ and $\|dC\| \neq 0$,*

$$\|m dC\| \neq 0, \quad (17)$$

the continuum field strength is nonzero in norm.

Reading. In a normed space, a vector is zero iff its norm is zero; combined with the fact that a product of nonzero reals is nonzero, the theorem gives the strict form of CA9: the nuclear channel is nonzero *in norm*, not merely as a formula. The numbers again: $|m| > 0$ and $\|dC\| > 0$ multiply to a positive norm. Formalized as `nuclear_channel_norm_ne_zero` in `YangMillsStrict.lean`.

b. What remains open. The topological limit of the lattice spacing to zero — the definition of a lattice field sequence converging to a continuum field, and the proof that the commutator field strength converges to the continuum $[A_\mu, A_\nu]$ — is the one layer not yet formalized. It is stated, not blurred. The strict estimates above are the norm prerequisites of that limit; the limit itself, and the matrix-valued generalization of the derivative theorem L2, are the open items named in Sec. X.

VIII. THE ASSEMBLED CHAIN

The full conditional chain:

$$\begin{aligned}
&\text{accelerated charge} \Rightarrow \text{field varies (CR1)} \\
&\Rightarrow \text{nuclear channel } m\dot{C} \neq 0 \text{ (CR2)} \\
&\Rightarrow \text{self-flattening sink (CR9, YM1)} \\
&\Rightarrow \text{sink noncommutativity (YM2) = self-interaction } [A, A] \text{ (CA3–CA4)} \\
&\Rightarrow \text{continuum mechanism (CA8–CA10)} \Rightarrow \text{mass gap (MG2b, CA7)}.
\end{aligned} \tag{18}$$

Every implication is a theorem in Lean 4; the chain is conditional only in the sense that its inputs — the flattening power, the contraction rate λ , the strand mass M_0 — are named and not derived. The two names of the Yang–Mills problem, *discreteness* and *self-interaction*, are here one mechanism: the mass gap is the discreteness of the strand count, and the self-interaction is the noncommutativity of the count-carrying sink operators. The chain reads as a single sentence: accelerated charge makes the field vary; a varying field is a self-flattening sink; the sink does not commute with itself across positions; that noncommutativity is the field’s self-interaction; and a self-interacting field whose mass is a count has a mass gap.

a. Relation to existing physics. The framework is deliberately conservative where standard physics is well tested. It keeps the Yang–Mills field strength (1) as the target structure: the commutator $[A_\mu, A_\nu]$ is what the lattice field strength is built to reproduce [2, 13]. It keeps the lattice method as the route from discrete to continuous, in the tradition of lattice gauge theory [3, 4]. It keeps the Einstein field equations at the level of their weak-field content (gravitation is the non-uniformity of the flow) and the Maxwell equations as the kinematics of the flow [14–16]. In these places the framework *rewrites the ontology without*

touching the numerics: the same equations, a different primitive. In two places it goes further and contradicts the standard ontology directly. (i) The mass gap: standard QFT takes it as a non-perturbative fact to be measured or simulated; here it is a theorem of the discreteness of mass. (ii) The self-interaction: standard QFT postulates $[A, A]$ by gauge invariance; here it is a theorem about the noncommutativity of sink contractions, and the two facts — gap and self-interaction — are identified as one mechanism. What is new, and flagged as such, is that identification itself.

IX. HONEST ASSESSMENT

a. Claimed. (i) Discreteness of mass (Postulate II.2) implies the empty interval $(0, M_0^2)$ (MG1–MG3): the electron–photon transition is an integer jump. The repository’s glueball mass squared is an integer, 0 or ≥ 1 (MG4). (ii) Sink contractions are noncommutative (YM2) with an exact commutator formula (YM3) and an exact magnitude formula (YM3b); the sink is a self-flattening dynamics with exponential decay (YM1). (iii) On a lattice, a color-matrix-valued gauge field has field strength equal to the nearest-neighbor commutator (CA3–CA4), the lattice avatar of $[A, A]$, and a constant color cycle has vanishing field strength (CA5). (iv) The four-force derivation is an existence mechanism for the continuum limit (CA8–CA10), with strict norm bounds and norm-nonzero estimates (L1–L3). (v) The chain from accelerated charge to the mass gap is a conditional sequence of theorems, assembled in CA7/CA10. All statements are formalized in Lean 4 with zero **sorry**.

b. Not claimed. (i) The Yang–Mills existence problem is not solved: the lattice construction is an algebraic skeleton, not a continuum four-dimensional quantum field theory; there is no path integral, no renormalization, no continuum spacetime. (ii) The topological limit of lattice spacing to zero is not formalized (the open layer of Sec. VII); in particular, the convergence of the discrete commutator $[A(t), A(t+1)]$ to the continuum $[A_\mu, A_\nu]$ is not proved. (iii) The numerical values λ , M_0 , and the flattening power are inputs, not derived; the framework therefore does not predict the scale of the QCD gap (about 1.6 GeV) — it predicts the shape of the spectrum, not its calibration. (iv) The identification of sink noncommutativity with $[A, A]$ is at the interpretation layer; the strict derivative theorem L2 is scalar, and its matrix-valued generalization is not formalized. (v) The lattice

is one-dimensional; no claim is made for 3+1-dimensional gauge theory on this basis.

c. Death conditions. The program dies if (a) a physical self-interaction is found with no discrete counterpart in this framework, or (b) the lattice field strength fails to converge to the continuum $[A_\mu, A_\nu]$ as the lattice spacing goes to zero, or (c) a noncommutative self-interaction is exhibited with no mass lower bound. These are falsifiable in principle, which is the point: a framework that cannot die cannot be tested.

X. CONCLUSION

The Yang–Mills problem names two facts: a positive mass gap and a non-abelian self-interaction. In the flowing-space framework these are not two facts but one. Mass is an integer count; the mass gap is the discreteness of that count — the interval $(0, M_0^2)$ is empty because there is no integer between 0 and 1. The sink is a self-flattening operator; its noncommutativity is the self-interaction — the exact commutator $\lambda^2(v_q - v_p)$ is the same shape as $[A, A]$, field-dependent and noncommutative. The chain that the preceding paper built from accelerated charge to the photon is here extended from the photon to the mass gap, through the self-interaction, with the continuum mechanism supplied by the framework’s own four-force derivation and bounded by strict estimates with the explicit factor 6.

The open items assemble into a short list of problems in the spirit of Hilbert’s program [17]: (i) derive the strand mass M_0 from the flow dynamics, turning the lattice glueball scale of about 1.6 GeV into a target rather than an import; (ii) derive the contraction rate λ and the flattening power from the divergence dynamics; (iii) formalize the topological limit — the convergence of lattice field sequences and of the commutator field strength to the continuum $[A_\mu, A_\nu]$; (iv) promote the strict derivative L2 to the matrix-valued setting, so that the four-force derivative is a theorem in the color space itself; (v) extend the $\mathbb{Z}$ lattice to 3+1 dimensions and test the chain against the empirical glueball spectrum [3–5]; (vi) find a physical observable for the sink noncommutativity $\lambda^2(v_q - v_p)$ — the framework’s candidate signature of self-interaction. Each is a well-posed question that this paper leaves with the boundary stated rather than blurred. The road is followed precisely; the mountain

is climbed only a little, just enough to be sure it is a mountain [8, 17].

- [1] A. Jaffe and E. Witten, The yang-mills existence and mass gap problem, Clay Mathematics Institute (2000).
- [2] C. N. Yang and R. L. Mills, Conservation of isotopic spin and isotopic gauge invariance, *Physical Review* **96**, 191 (1954).
- [3] C. J. Morningstar and M. Peardon, Glueball spectrum from an anisotropic lattice study, *Physical Review D* **60**, 034509 (1999).
- [4] Y. Chen, A. Alexandru, S.-J. Dong, T. Draper, I. Horvath, F. Lee, K. Liu, N. Mathur, C. Morningstar, M. Peardon, *et al.*, Glueball spectrum and matrix elements on anisotropic lattices, *Physical Review D—Particles, Fields, Gravitation, and Cosmology* **73**, 014516 (2006).
- [5] B. Collaboration *et al.*, Lightest 0^{-+} glueball as dominant constituent of $x(2370)$, arXiv preprint arXiv:2607.20366 (2026).
- [6] S. Weinberg, *The quantum theory of fields*, Vol. 2 (Cambridge university press, 1995).
- [7] L. De Moura, S. Kong, J. Avigad, F. Van Doorn, and J. von Raumer, The lean theorem prover (system description), in *International conference on automated deduction* (Springer, 2015) pp. 378–388.
- [8] Y. Liu, ProjectionPhysics — Algebraic Emergent Physics (Lean 4 formalization and verification scripts), <https://github.com/logos-42/Hibs-Physics> (2026).
- [9] A. J. Hamilton and J. P. Lisle, The river model of black holes, *American Journal of Physics* **76**, 519 (2008).
- [10] Y. Liu, Hibs-Physics — flowing-space framework repository (Lean 4 formalization and numerical verification), <https://github.com/logos-42/Hibs-Physics> (2026).
- [11] B. Riemann, Über die anzahl der primzahlen unter einer gegebenen grösse, *Monatsberichte der Berliner Akademie* (1859).
- [12] Y. Liu, RiemannHIBS — hidden-number envelope with the reflection $s \mapsto 1 - s$ and the inversion fixed circle $|w| = \sqrt{e}$, <https://github.com/logos-42/RiemannHIBS> (2026).
- [13] M. E. Peskin and D. V. Schroeder, *An introduction to quantum field theory* (Westview Press, 1995).
- [14] A. Einstein, Die feldgleichungen der gravitation, *Sitzungsberichte der Königlich Preußischen*

- Akademie der Wissenschaften , 844 (1915).
- [15] W. Gordon, Zur lichtfortpflanzung nach der relativitätstheorie, Annalen der Physik **377**, 421 (1923).
- [16] J. C. Maxwell, *A dynamical theory of the electromagnetic field* (Wipf and Stock publishers, 1996).
- [17] D. Hilbert, Mathematische probleme, Göttinger Nachrichten **3**, 253 (1900).